

AMENDMENT NO. 1 JUNE 2024

TO

IS 9079 : 2018 MONOSET PUMPS FOR CLEAR, COLD WATER FOR AGRICULTURAL AND WATER SUPPLY PURPOSES — SPECIFICATION

(Third Revision)

(Second cover, foreword) — Insert the following after second paragraph:

‘Graphs for calculating minimum pump efficiency are replaced by equation (1). Equation (1) is adopted from commission regulation (EU) No. 547/2012 with regarding to eco-design requirements for water pumps for minimum efficiency index (MEI), for pump type end suction close coupled (ESCC).’

(Page 2, clause 9.2) — Substitute the following for the existing:

‘9.2 Voltage and Frequency Variation

Monoset shall be capable to operate in the following range of voltage and supply frequency:

- a) The terminal voltage differing from this rated value by not more + 6 percent and – 15 percent;
- b) The frequency differing from its rated value by not more than ± 3 percent; and
- c) Any combination of above.’

(Page 2, clause 9.4) — Substitute the following for the existing:

‘9.4 Efficiency

The pump efficiency of monoset pumps at duty point as declared by the manufacturer for 2 and 4 poles shall not be less than that calculated from 13.4.1.’

(Page 5, clause 13.1.2, line 4) — Substitute ‘However, after applying the tolerance, overall efficiency value shall not be less than that obtained after multiplying the motor efficiency factor and pump efficiency as derived from equation (1) of 13.4.1.’ for ‘However, after applying the tolerance, overall efficiency value shall not be less than that derived from Fig. 3 to Fig. 6.’

(Page 6, clause 13.4.1) — Substitute the following for the existing:

‘13.4.1 Pump Efficiency

Pump efficiency shall be calculated using the equation as given below, and efficiency corresponding to MEL 0.2 shall be taken as the minimum pump efficiency:

$$\eta_p = 88.59 x + 13.46 y - 11.48 x^2 - 0.85 y^2 - 0.38 xy - C \dots(1)$$

where

- η_p = minimum pump efficiency;
- x = $\ln(n_s)$;
- y = $\ln(Q)$;
- $\ln$ = natural logarithm;
- n_s = specific speed; and
- Q = discharge rate in m^3/h ; and
- C = values for different pump efficiency levels.

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$$n_s = n \cdot \frac{\sqrt{Q/3600}}{(H)^{0.75}}$$

where

n = 'rotational speed' in revolutions per minute (rpm);

Q = 'discharge rate' in m³/h;

H = 'head' per stage in meters (m); and

C = values for different pump efficiency levels.

Minimum Efficiency Level, MEL		0.20	0.30	0.40	0.50	0.60	0.70
C	2 Pole, speed	133.82	132.23	130.77	129.86	128.8	127.75
	4 Pole, speed	131.2	129.77	128.46	127.38	126.57	125.46

NOTES

1 Other MEL values are for the guidance of the manufacturer and to upgrade to higher values of pump efficiency.

2 An illustrative example for calculating pump efficiency using above equation is given in Annex B.

3 The calculated value for specific speed and minimum pump efficiency shall be rounded off up to 2 decimal places.'

(Page 6, clause 13.4.2) — Substitute the following for the existing:

'13.4.2 For minimum overall efficiency of the pumpset, multiply minimum pump efficiency calculated as per 13.4.1 by corresponding motor efficiency factor as per Table 5 for three phase motors and as per Tables 6 to 8 for single phase motors.'

(Page 6, clause 13.4.4) — Delete '13.4.4'.

(Page 12, Fig. 3) — Delete.

(Page 12, Fig. 4) — Delete.

(Page 13, Fig. 5) — Delete.

(Page 13, Fig. 6) — Delete.

(Page 14, Annex A) — Insert the following the end:

‘ANNEX B

(Clause 13.4.1)

EXAMPLE OF MINIMUM PUMP EFFICIENCY CALCULATION FOR END SUCTION VOLUTE TYPE MONOSET (2 POLE) DUTY POINT

B-1 INPUT DATA

Head = 22 m

Discharge rate = 11 lps

Speed = 2 900 rpm

B-2 CALCULATIONS

B-2.1 To Find out Discharge Rate, Q, in m³/h

Discharge rate = 11 lps × 3.6 = 39.6 m³/h.

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B-2.2 To Find out Specific Speed, n_s

$$n_s = n \cdot \frac{\sqrt{Q/3600}}{(H)^{0.75}}$$

$$n_s = 2900 \times \frac{\sqrt{39.6/3600}}{(22)^{0.75}} = 29.94$$

NOTE — For specific speed calculation we have to use discharge (Q), in m³/h.

B-2.3 To Find out Minimum Pump Efficiency, η_p

$$\eta_p = 88.59 x + 13.46 y - 11.48 x^2 - 0.85 y^2 - 0.38 xy - C \dots (1)$$

$$x = \ln(n_s) = \ln(29.94) = 3.39$$

$$y = \ln(Q) = \ln(39.60) = 3.67$$

NOTE — For minimum pump efficiency calculation we have to use discharge (Q), in m³/h.'

C = 133.82 (C-value for MEL 0.2 to be taken from the table for minimum efficiency calculation for 2 pole).

$$\eta_p = 88.59 \times 3.39 + 13.46 \times 3.67 - 11.48 \times 3.39^2 - 0.85 \times 3.67^2 - 0.38 \times 3.39 \times 3.67 - 133.82$$

$$\eta_p = 67.79 \% \text{ (Minimum pump efficiency).}$$

(MED 20)

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